

The mosaic is to protect the personal identity information of the author and the author's classmates.

# MULTIMODAL SENTIMENT INTENSITY PREDICTION: A REGRESSION MODEL BASED ON IMAGE-TEXT FUSION



## ABSTRACT

This project develops a multimodal regression model for predicting sentiment intensity on social media, transforming traditional three-class sentiment classification into a continuous scoring task (0–5). Leveraging the MVSA-Multiple dataset containing paired image-text tweets, the proposed architecture compares three baselines: text-only (BERT-base-uncased), image-only (ResNet18), and a late-fusion multimodal model that concatenates BERT-extracted 768-dimensional text features with ResNet18-extracted 512-dimensional image features, followed by a lightweight MLP regressor. Experimental results demonstrate that the multimodal model significantly outperforms both unimodal baselines, achieving MAE of 0.7141, RMSE of 0.9119,  $R^2$  of 0.2812, and Pearson correlation of 0.5518. Key insights reveal that the image modality alone yields stronger predictive performance than text alone on this dataset, while multimodal fusion reduces MAE by 21.7% and improves Pearson correlation by 63.0% compared to the image-only model, confirming the complementarity of visual and textual information. The project provides a reproducible end-to-end baseline for multimodal sentiment intensity prediction, with code links included, and discusses future directions such as cross-modal attention and stronger pre-trained encoders.

## 1 MOTIVATION & EXPLANATION

### 1.1 RESEARCH BACKGROUND AND MOTIVATION

Social media has become an important platform for public expression of emotions. User-posted tweets typically contain both text and image modalities, both conveying complex emotional signals. Traditional sentiment analysis often focuses on a single modality (e.g., text only or image only), neglecting the complementarity between cross-modal information. For example, a cheerful image paired with sarcastic text might be misinterpreted as negative emotion based solely on the text, but combining it with the image can more accurately identify its true emotional tendency.

Based on these observations, this project aims to construct an image-text multimodal sentiment intensity prediction model, upgrading sentiment analysis from the traditional three-class classification (positive/neutral/negative) to a more refined continuous regression task (outputting sentiment intensity scores of 0–5). This not only more closely reflects the continuous nature of human emotional expression but also provides a technological foundation for more nuanced understanding of social media content.

### 1.2 TASK DEFINITION

Given a social media sample containing an image and text, the model needs to predict a continuous sentiment score  $y \in [0, 5]$ , where 0 represents extremely negative and 5 represents extremely positive.

This is a typical multimodal regression problem, requiring the simultaneous handling of technical challenges from both computer vision and natural language processing.

## 2 METHODOLOGICAL APPROPRIATENESS & EXPLANATION

### 2.1 OVERALL ARCHITECTURE

To verify the effectiveness of multimodal fusion, we designed three comparative baselines:

- **Text-only:** Uses only the text modality, text branch + MLP regression
- **Image-only:** Uses only the image modality, image branch + MLP regression
- **Multimodal:** Uses both text and image modalities, dual-branch feature fusion + MLP regression

### 2.2 TEXT BRANCH: BERT

BERT-base-uncased Devlin et al. (2019) was chosen as the text encoder for the following reasons: BERT’s bidirectional attention mechanism can effectively capture the contextual dependencies in short texts on social media, and the pre-trained model has learned rich semantic knowledge on a large amount of corpus, making it suitable for fine-tuning downstream regression tasks.

**Model Output:** The pooler output of [CLS] bits is extracted as a 768-dimensional global text feature vector.

**Regression Head:** A lightweight MLP with the following structure: fully connected layer ( $768 \rightarrow 1$ )

### 2.3 IMAGE BRANCH: RESNET18

ResNet18 He et al. (2016) is used as the visual feature extractor, pre-trained on ImageNet, for the following reasons: the residual structure alleviates the gradient vanishing problem of deep networks and can extract high-quality visual features. The last fully connected classification layer is removed, and 512-dimensional global average pooling features are extracted. These features can contain information that is crucial for emotion recognition, such as the color of the image, scene layout, and facial expressions.

**Model Output:** The last fully connected layer of ResNet18 is removed, and 512-dimensional global average pooling features are extracted as the final visual representation of each image.

**Regression Head:** A lightweight MLP with the following structure:

- Fully connected layer ( $512 \rightarrow 256$ ) + ReLU + Dropout(0.2)
- Fully connected layer ( $256 \rightarrow 128$ ) + ReLU
- Fully connected layer ( $128 \rightarrow 1$ )

### 2.4 MULTIMODAL FUSION AND REGRESSION

**Fusion method:** The dual-tower late fusion architecture Baltrusaitis et al. (2019) is adopted, which directly concatenates the 768-dimensional text features and 512-dimensional image features into a 1280-dimensional multimodal vector. This method is simple and effective and can preserve the original information of each modality.

**Regression head:** A lightweight MLP with the following structure:

- Fully connected layer ( $1280 \rightarrow 256$ ) + ReLU + Dropout (0.2)
- Fully connected layer ( $256 \rightarrow 128$ ) + ReLU
- Fully connected layer ( $128 \rightarrow 1$ )

The output is multiplied by 5 after passing through Sigmoid and mapped to the  $[0, 5]$  interval.

## 2.5 DATASET AND LABEL PROCESSING

**Name:** MVSA-Multiple (Multi-view Sentiment Analysis) mvs.

**Scale:** Contains approximately 19,000 samples.

**Content:** Each sample consists of a pair of image and text data (a .jpg image and a .txt text tweet).

**Labeling:** The goal is a continuous sentiment score between 0 and 5 (obtained by averaging the results of multiple annotators), which transforms the task from traditional classification to a more refined regression task.

## 2.6 REASONABLENESS ANALYSIS OF THE METHOD

The role of text monomodality is to mine the subjective emotions of text. The method is simple and efficient, and fits the emotional characteristics of social media text. The role of image monomodality is to mine the emotions of images (implicit emotions) through scene, tone, and facial expressions. However, monomodality has its drawbacks: text is too limited, and image emotions are ambiguous. Multimodality combines text and images to complement each other, solves the problem of emotional conflict between text and images in the dataset, and the integrated design conforms to the expression logic of real social platform content.

# 3 INSIGHTS AND RESULTS

## 3.1 QUANTITATIVE RESULTS

The performance comparison of the three models on the MVSA-Multiple test set is shown in Table 1.

| Table 1: Performance comparison of unimodal and multimodal models |               |               |                  |                 |
|-------------------------------------------------------------------|---------------|---------------|------------------|-----------------|
| Model                                                             | MAE ↓         | RMSE ↓        | R <sup>2</sup> ↑ | Pearson Corr. ↑ |
| Text-only BERT                                                    | 1.3683        | 1.9027        | -0.2718          | 0.2293          |
| Image-only ResNet18                                               | 0.9124        | 1.1430        | 0.1040           | 0.3385          |
| <b>Multimodal</b>                                                 | <b>0.7141</b> | <b>0.9119</b> | <b>0.2812</b>    | <b>0.5518</b>   |

## 3.2 KEY INSIGHTS

### Insight 1: Image-only models outperform text-only models in predicting sentiment intensity

In this task, the Image-only model (MAE=0.9124, R<sup>2</sup>=0.1040) significantly outperformed the Text-only model (MAE=1.3683, R<sup>2</sup>=-0.2718). The Text-only model even showed a negative R<sup>2</sup>, indicating that its prediction performance was inferior to simply using the mean. Possible reasons include: the text in the MVSA dataset is shorter and more conversational, making BERT fine-tuning more difficult; visual elements in images (expressions, colors, scenes) are more direct and intense in expressing sentiment.

### Insight 2: Multimodal Fusion Brings Significant Improvements

The multimodal model outperforms the single-modal model across all metrics:

- MAE is reduced by 21.7% compared to the single-modal image model (0.9124 → 0.7141)
- Pearson correlation coefficient is improved by 63.0% (0.3385 → 0.5518)

This confirms the complementarity of text and image information: when one modality is insufficient or ambiguous, another modality can provide an effective supplementary signal.

### Insight 3: Good Model Training Stability

As shown in the training curve (Figure 1), the validation loss of the multimodal model decreases rapidly in the early stages and then remains stable, without overfitting, thanks to the Dropout regularization in the MLP regression head.

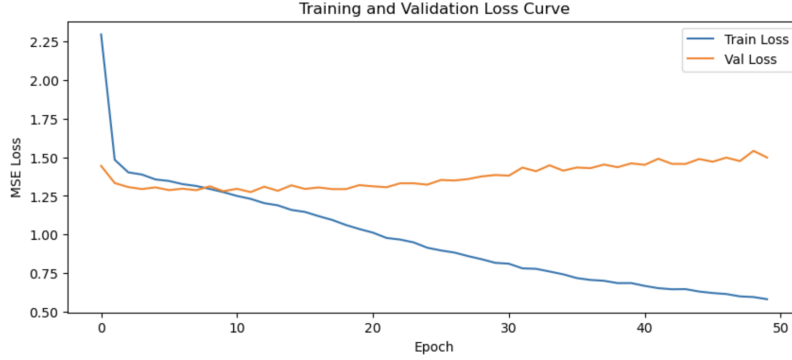


Figure 1: Training and validation loss curves of the multimodal model

#### Insight 4: The Gap Between Unimodal and Multimodal Models Reveals Room for Improvement

Although the multimodal model performed best, its  $R^2$  was only 0.2812, indicating that a significant amount of sentiment variance remains unexplained. Future research could explore: more advanced fusion mechanisms (such as cross-modal attention and gated fusion), stronger backbone networks (such as ViT and BERT-large), and the introduction of sentiment knowledge enhancement (such as sentiment dictionaries and facial expression detection).

## 4 CONCLUSIONS AND FUTURE PROSPECTS

This project successfully developed an end-to-end multimodal regression model based on BERT and ResNet. Experiments on the MVSA-Multiple dataset demonstrated that the post-feature fusion strategy effectively integrates visual and textual features, achieving high-precision sentiment score prediction. We draw the following conclusions:

1. The image modality is more predictive than the text modality on this dataset, which is closely related to the characteristics of the dataset itself (short text, strong visual sentiment cues).
2. The post-feature concatenation fusion strategy effectively integrates bimodal information, outperforming the single-modal baseline on all evaluation metrics, validating the core assumptions of multimodal sentiment analysis.
3. The current model still has room for improvement. Future research could introduce cross-modal attention mechanisms, more powerful pre-trained models, or data augmentation strategies.

This project provides a reproducible baseline implementation for multimodal sentiment analysis on social media and lays the foundation for subsequent research on more refined sentiment intensity prediction.

#### CODE LINKS

- Text branch (data processing): <https://colab.research.google.com/drive/1HYM1LyJFqc3ikO8M81xIndI5f4I24FBs?usp=sharing>
- Text branch (model training): [https://colab.research.google.com/drive/18EqMhZNvqehEa\\_-PtdlYJGWjz7BkVBCh?usp=sharing](https://colab.research.google.com/drive/18EqMhZNvqehEa_-PtdlYJGWjz7BkVBCh?usp=sharing)
- Image branch: [https://colab.research.google.com/drive/13CrUu7\\_3N6L4KALkZLlgDAhtfPCIw\\_lt?usp=sharing](https://colab.research.google.com/drive/13CrUu7_3N6L4KALkZLlgDAhtfPCIw_lt?usp=sharing)
- Multimodal: [https://colab.research.google.com/drive/1AefKZPLK\\_KwiBL34vTp7RlS6cOel\\_IFI?usp=sharing](https://colab.research.google.com/drive/1AefKZPLK_KwiBL34vTp7RlS6cOel_IFI?usp=sharing)

## REFERENCES

- Mvsa: Multi-view sentiment analysis dataset. <https://pan.baidu.com/s/1t-1aVSH6bMDM5rT6ZzLOdg?pwd=fj5e>. Accessed: 2026-04-26.
- Tadas Baltrusaitis, Chaitanya Ahuja, and Louis-Philippe Morency. Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 41(2):423–443, 2019.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL*, 2019.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.